

Sai Pavan Kumar Kasaragadda | Full Stack Developer

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SUMMARY

Full Stack Java Developer with 5 years of experience designing and building scalable web applications, APIs, and microservices using Java (Spring Boot, Hibernate) and React.js across insurance, healthcare, and FinTech domains. Proficient in cloud platforms (AWS, Azure), databases (PostgreSQL, MongoDB), and DevOps automation (Docker, Kubernetes, Jenkins). Experienced in integrating AI/ML services and event-driven systems into enterprise platforms. Adept at applying OOP, design patterns, and agile methodologies to deliver secure, high-performing, and maintainable digital solutions.

SKILLS

Programming & Scripting: Frontend Development:

Java, JavaScript (ES6+), TypeScript, Python, SQL, HTML5, CSS3, Bash/Shell Scripting
React.js, Redux, Next.js, Angular.js, Vue.js, Material UI, Tailwind CSS, Responsive Design, Component-Based Architecture

Backend & API Development:

Spring Boot, Spring MVC, Hibernate, JPA, Node.js, Redis, RESTful APIs, GraphQL, Microservices, Swagger/OpenAPI, SOAP, FHIR, HL7 V2

Cloud, DevOps & CI/CD:

AWS (EC2, ECS, Lambda, RDS, S3), Azure DevOps, GCP, Docker, Kubernetes, Jenkins, GitHub Actions, Terraform, Ansible, Infrastructure-as-Code (IaC), CI/CD Pipelines

Databases & Messaging Systems:

PostgreSQL, MySQL, MongoDB, Redis, Oracle, Apache Kafka, RabbitMQ, Data Modeling, Query Optimization

Testing, Security & Methodologies:

JUnit, Mockito, Selenium, Postman, TDD, SonarQube, OAuth2, JWT, Agile, Scrum, JIRA, OOP, Design Patterns, ELK, WebSockets

AI & Integration Tools:

Palantir Foundry, LangChain, AWS Bedrock, GenAI, API Gateway, Event-Driven Architecture, Digital Twin Concepts, OpenAI API, Claude, Prompt Engineering, RAG

EXPERIENCE

AIG, USA | Full Stack Developer

Jul 2023 – Current

- Developed and enhanced Java Spring Boot microservices integrated with React.js frontend for AIG's GenAI-powered underwriting assistant, improving data accuracy from 75% to 90%.
- Designed and deployed RESTful APIs to connect Palantir Foundry with insurance platforms, automating real-time exchange of 125+ key data elements across underwriting and claims.
- Implemented secure API authentication (OAuth2, JWT) and modular UI components to comply with NAIC AI governance and enterprise security standards.
- Optimized PostgreSQL queries and schema designs for real-time analytics and underwriting data models, reducing query latency by 40%.
- Deployed containerized Spring Boot services using Docker, AWS ECS, and Jenkins CI/CD pipelines, improving release reliability and system uptime.
- Partnered with AI/ML teams to embed predictive analytics APIs within underwriting workflows, accelerating AI adoption across AIG's digital ecosystem.

AIG, USA | Full Stack Developer Intern

Mar 2023 – May 2023

- Assisted in building UI components using React.js and integrated them with Spring Boot REST APIs, improving underwriting workflow efficiency by 12%.
- Supported the development of microservices connecting Palantir Foundry with core insurance systems for seamless policy and claims data exchange.
- Wrote JUnit and Postman test cases for end-to-end validation of AI-driven underwriting modules, ensuring compliance and data consistency.
- Collaborated with backend engineers to optimize SQL queries and API response times, gaining hands-on experience with large-scale, cloud-hosted insurance systems.

CitiusTech, India | Full Stack Java Developer

Jan 2020 – Nov 2021

- Built and maintained healthcare web applications using Java (Spring Boot, Hibernate) and React.js, enabling real-time visualization of patient and claims data.
- Integrated FHIR and HL7 V2 APIs to connect EHR systems like Epic with third-party analytics platforms, reducing data transfer latency by 40%.
- Automated ETL data workflows on AWS (Lambda, S3, Aurora PostgreSQL), improving operational efficiency by 50% and ensuring HIPAA compliance.
- Designed and tuned PostgreSQL and MongoDB schemas to enhance reporting and analytical performance by 45%.
- Containerized applications using Docker and automated deployments via Jenkins pipelines, achieving 35% faster release cycles.
- Worked closely with clinical SMEs and data engineers to integrate AI/ML insights into the application UI for better care management decision-making.

EDUCATION

Masters in Computer Science | University Of New Haven, CT

May 2023

Bachelors in Computer Science & Engineering | Bapatla Engineering College, INDIA

Jul 2021

PROJECTS

AI-Augmented Kubernetes Monitoring

- Built a monitoring pipeline using Prometheus + Grafana ML to detect anomalies in pod CPU/memory trends, reducing false alerts by 40%.

AI ChatOps for DevOps –

- Integrated a GenAI-powered chatbot into Slack to automate on-demand AWS resource provisioning and log searches, reducing manual intervention for routine tasks.

2048 Game Deployment

- Containerized a web-based 2048 puzzle game and deployed it to Kubernetes clusters, leveraging Helm charts for version-controlled, repeatable deployments.
- Containerized web app with Helm, Jenkins CI/CD, and Grafana monitoring – reduced deployment time by 30% (used for team training).
- Automated build and deployment using Jenkins, reducing release time by 30% and ensuring consistent multi-environment delivery.
- Configured Grafana dashboards for live game performance metrics, enabling proactive resource scaling during load spikes.

Vision-Based Vehicle Detection in Traffic Surveillance System

- Developed a computer vision pipeline using OpenCV & YOLO to detect and classify vehicles from real-time traffic camera feeds, enabling data collection for congestion analytics.
- Deployed model inference service in Docker containers with GPU acceleration, reducing detection latency by 40%.
- Integrated AWS S3 & Lambda for automated image storage and processing, enabling scalable traffic data analysis workflows.

Mario Kantra Game (Masters Project)

- Designed and implemented a 2D Mario-inspired platformer using Java & LibGDX, integrating collision detection, scoring systems, and level progression.
- Built a Spring Boot + React e-commerce demo with Kafka-based order processing, Stripe payment integration, and PostgreSQL for inventory management. Deployed on AWS ECS with auto-scaling.
- Packaged the game into an executable build and hosted it on AWS S3 for public access, demonstrating CI/CD for game deployment.
- Added AI-assisted gameplay testing using scripted bots to validate collision handling and level difficulty balancing before release.

CERTIFICATIONS

AWS Certified Solutions Architect – Associate

AWS Educate Introduction to Generative AI - Training Badge